

# Bryan Heddle

Hamilton, Ontario

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## EDUCATION

University of Western Ontario

Mechatronics with AI Systems Engineering

Sept 2022 – May 2028 | GPA: 3.7 / 4.0

London, Ontario

## TECHNICAL SKILLS

**Languages:** Python, C, C++, SQL, Bash, JavaScript, HTML/CSS

**ML & CV:** PyTorch, TensorFlow, Keras, scikit-learn, Hugging Face, OpenCV, RL, pandas, NumPy

**Robotics, Embedded & Systems:** ROS 2, NVIDIA Isaac Sim/Lab, CUDA, CAN, Docker, AWS, Linux, Git/GitLab CI

## WORK EXPERIENCE

Computer Vision Researcher | *VIP Lab, University of Waterloo*

May 2026 – Aug 2026

- Cut median reprojection error  $\sim 92\%$  (**15.1 px**  $\rightarrow$  **1.2 px**) vs DJI vendor poses by matching RGB drone imagery to 3D LiDAR with SIFT, rejecting outliers with RANSAC, then solving PnP for metric camera poses.
- Built a **CuPy** CUDA LiDAR renderer that chunk-streams and splat-renders  $\sim 600\text{M}$ -point Terra clouds with on-device z-buffering, enabling full-scene pose refinement without OOM.
- Reduced point-to-mesh error from DJI Terra **4.7 cm** to **0.8 cm** with a Poisson LiDAR mesh that removed surface drift.

Embedded Developer Intern | *Geotab*

May 2025 – May 2026

- Architected a **USB/IP** peripheral migration for Nordic nRF dongles onto HIL rigs, patching and compiling the **Linux vhci driver** to free **26** server USB ports and scale virtual controller support to **128+**.
- Owned end-to-end bring-up of **14 test rigs** gating firmware test access to **4** new products, handling everything from custom PCB wiring and Ansible deployment to GitLab runner registration and oscilloscope debugging.
- Eliminated GPIO polling timeouts on smoke tests by replacing a server-side poll loop with event-driven **GPIOZero** callbacks, cutting server CPU load and delivering consistent smoke-test results across concurrently tested devices.
- Engineered a Python CI service that parses test reports and publishes hierarchical **Google Chat** cards (pipeline, job, and per-test detail) across all HIL products to **140+** engineers.
- Integrated a custom **MCP** server that dynamically queries **GitLab** job logs and source files based on failure context, feeding retrieved artifacts to an AI model to generate per-test root-cause summaries on each job card.

## DESIGN TEAM

President & Lead Software Engineer

Sept 2023 – Present

*Western Engineering Mars Rover Team*

London, Ontario

- Cut rover teleop latency from **240 ms** to **44 ms** by rebuilding the WebRTC stack in C++ with **libdatachannel**, designing a zero-copy capture-encode-packetize pipeline optimized for minimal delay over quality.
- Designed a multi-node **CAN** network across **ESP32 (TWAI)** and **Jetson (SocketCAN)**, tuning per-message frequencies across **8** motors to **44%** bus utilization and exposing all devices to **ROS 2** via a hardware abstraction layer.
- Implemented a C++ **ros2\_control** swerve IK plugin mapping `/cmd_vel` to 8 joints with shortest-path 180° wrap.

## RESEARCH & PROJECTS

ServoSim | **PI: Dr. Elvis Chen**

- Authored and won a **\$15K NVIDIA Academic Grant** (RTX PRO 6000, Jetson AGX Orin) for sim-to-real research.
- Built a characterization rig collecting synchronized commanded, reported, and ground-truth encoder positions on **STS3235** and **STS3250** servos, generating a  $\sim 10\text{GB}$  open-source dataset for actuator net training.
- Developing **ServoSim**, an **Isaac Lab** extension introducing **MLP** and **LSTM** position-based actuator classes with hardware-calibrated domain randomization for serial-bus servos.

ERGOS: Dynamic Humanoid Robot | GitHub

- Raised **\$5K+** in sponsorships by pitching ERGOS as a long-term humanoid robotics research platform.
- Designed and built a **20-DOF** belt-driven humanoid from scratch in **SolidWorks**, with a balanced COM, custom **Altium** PCB, embedded electronics, and control systems.
- Implemented half-duplex TTL servo communication using gate buffers and GPIO direction switching, forking the Feetech serial library to enforce precise timing after buffer drain.
- Fused **IMU** data over **I2C** with a Kalman filter for stabilized orientation estimates used in balance control.